

Program: Diploma in Integrated Circuit Design and Fabrication	
Course Code : 3377	Course Title: Electric Circuits and Instrumentation
Semester : 3	Credits: 0
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To develop the skill to diagnose and rectify the electric circuit networks
- To provide learning experience on characteristics of testing and measuring instruments.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic Engineering Mathematics principles and theorems	1002 2002	Mathematics I & I	1 & 2
Knowledge of current, voltage & magnetism	1003 2003	Applied Physics I & II	1 & 2
Representation of AC signals, voltage and power	2031	Fundamentals of electrical and electronic engineering	2

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive level
CO1	Solve a given AC circuit to find various parameters using the concept of AC signals and the behavior of AC through various components.	13	Applying
CO2	Apply various network theorems for simplifying electric circuits and networks and illustrate the operations of transformers	15	Applying
CO3	Explain the working principle of various measuring Instruments	15	Understanding
CO4	Explain the working principle of CRT and CRO.	15	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Solve a given AC circuit to find various parameters using the concept of AC signals and the behavior of AC through various components.		
M1.01	Illustrate the concept of AC signals	2	Understanding
M1.02	Explain the behavior of AC through R-L-C components	3	Understanding
M1.03	Solve the power in given AC circuits by keeping the concept of phasor diagram.	4	Applying
M1.04	Solve given series and parallel RLC circuits and find the various parameters	4	Applying
Contents: Concept of alternating voltage and current - Different waveforms (Sine, Square, Triangular and Sawtooth) - representation of alternating quantities - define the terms cycle, time period, frequency, amplitude, phase, maximum value, rms value, average value form factor AC through resistance, inductance, and capacitance (Solve simple problems) - Phasor diagrams- power factor definition - calculation of active, reactive and apparent power, Resonance and Q factor in an RLC circuit - series and parallel AC circuits (simple problems).			
CO2	Apply various network theorems for simplifying electric circuits and networks and illustrate the operations of transformers.		
M2.01	Illustrate various network theorems	3	Understanding
M2.02	Apply various network theorems for solving electrical and electronics circuits.	6	Applying
M2.03	Explain principle and operations of transformers	3	Understanding
M2.04	List the types and applications of Transformers.	3	Remembering

	Series Test - I	1	
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Contents: Network Theorems and Transformers

Ohm's law - Kirchhoff's law – Mesh analysis – Node analysis- Superposition theorem - Thevenin's theorem - Maximum power transfer theorem (Solve simple problems)
Transformers: working principle of transformer - construction of transformer - elementary theory of an ideal transformer - voltage transformation ratio and rating of a transformer - emf equation derivation - losses in transformers - types, applications of transformers

CO3	Explain the working principle of various measuring instrument		
M3.01	Summarize specifications of measuring instruments.	3	Remembering
M3.02	Explain the working principle and conversion of galvanometer into Voltmeter and Ammeter	4	Understanding
M3.03	Explain PMMC and MI instruments	4	Understanding
M3.04	Explain the working of analog and digital meters.	4	Understanding

Contents: Measurements and Measuring Instruments

Static characteristics of an instrument-Accuracy & Precision, Resolution, Expected value, Sensitivity. Types of Errors- static errors-gross errors, systematic errors, random errors
Construction and Working principle of Galvanometer and conversion into Ammeter & Voltmeter Permanent Magnet Moving Coil Instruments (PMMC) – construction and principle
Moving Iron type Instruments (MI)- working and construction- both attraction type and repulsion type
Electro Dynamo Type Instruments': construction and working; Analog- basic DC & AC voltmeter, Digital Voltmeter- ramp type, Digital Multimeters, Single-Phase Energy Meter

CO4	Explain the working principle of CRT and CRO.		
M4.01	Illustrate construction and operation of CRT	4	Understanding
M4.02	Explain working of CRO	4	Understanding
M4.03	Summarize measurements using CRO	3	Understanding
M4.04	Explain block diagram and features of DSO	4	Understanding
	Series Test – II	1	

Contents: Oscilloscopes

Cathode ray tube: construction, operation, screen materials, graticules, electrostatic focusing- Vertical deflection system, Horizontal deflection system, Delay line Block diagram and working of CRO, applications Multiple Trace CRO-definition, Dual trace Oscilloscope- block diagram and working Procedure for Measurements using CRO- Voltage (DC and AC), time delay, frequency and phase angle by using Lissajous patterns Digital Storage Oscilloscope- block diagram, features. Oscilloscope probes: 1:1 and 10:1 probe- equivalent circuit and description.

Text/Reference

T/R	Book Title/Author
T1	B. L. Thereja and A. K. Thereja, "Textbook of Electrical Technology: Part 1 - Basic Electrical Engineering in S. I. Units", S. Chand Publication, 2012
T2	Electronic Instrumentation - H S Kalsi
T3	Electronics Measurements And Instrumentation- U.A Bakshi and A.V. Bakshi
R1	B. L. Thereja and A. K. Thereja, "Textbook of Electrical Technology: Part 2-AC and DC machines", S. Chand Publication, 2012
R2	Dr Ganesh Rao & R Sreenivasa, "Network Analysis: A simplified approach", Cengage
R3	Electronic Instrument and measurement technique – W.D Cooper
R4	Electronic Measurement and Instrumentation – J.G JOSHI
R5	Electronics and Electrical Measurements and Instrumentation - J B Gupta - S K Kataria

Online resources:

Sl.No.	Website Link
1	https://www.khanacademy.org/electrical/
2	https://ocw.mit.edu/electrical
3	https://www.electrical4u.com/cathode-ray-oscilloscope-cro/
4	https://myclassbook.org/darsonval-meter-movement-principle/
5	https://www.tutorialspoint.com/electronic_measuring_instruments/electronic_measuring_instruments_transducers.htm